

CHATAKE INNOWORKS PVT. LTD.

MINDFORGEAI INTERNSHIP PROGRAM

Date: August 2026 **Document:** Project Abstract and Outline

CONSTRUCTVISION AI

Smart Residential Construction Inspection and Site Intelligence System

Particular	Details
Course title	Engineering Internship Project
Course number	INT-2026-01
Students	Ritika Bhumkar / Laiba Mulani
Student IDs	2026CE001 / 2026CE002
Degree programme	Diploma in Civil Engineering
Research area	Artificial Intelligence, Computer Vision and Civil Infrastructure Inspection
Project carried out at	Chatake Innworks Pvt. Ltd., MindForgeAI Division, Solapur, Maharashtra, India

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1. Broad Area of Work

ConstructVision AI belongs to the interdisciplinary areas of civil engineering, computer vision, digital-twin workflows, structural-health monitoring, Internet of Things (IoT) concepts and construction-data management. The project develops an evidence-first inspection workspace intended to organise site images, asset context, basic condition indicators, repair information and reports in one Streamlit application.

The current prototype supports image-based rule processing, a parametric three-dimensional structural view, synthetic telemetry, preliminary structural-screening calculations, repair-cost estimation, history and report workflows. It uses a rule-based OpenCV workflow to identify visible image features and estimates approximate dimensions through a user-supplied pixel-to-millimetre scale. The project also defines typed records for assets, captures, telemetry and future analysis requests, and it reports whether external reconstruction or solver workers are available.

The project does not claim that the current interface contains trained YOLO inference, calibrated photogrammetry, live production sensor ingestion, validated finite-element analysis, CFD, seismic time-history simulation or carbon forecasting. These are future extensions that require data, calibration, testing and qualified engineering review.

2. Background

Construction and infrastructure assets can develop cracks, spalling, corrosion, leakage, honeycombing and surface deterioration during their service life. Delayed identification can increase maintenance cost, interrupt operation and, in severe cases, create safety risks. Inspections are often documented across photographs, notebooks, spreadsheets and verbal site communication. This makes it difficult to trace an observation to a component, measurement method, recommended action and final report.

Manual inspection by qualified engineers remains essential, but it can be time-consuming and inconsistent when evidence is fragmented. Computer vision and structured digital records can assist by organising images, highlighting visible features and preserving preliminary results with their assumptions. IoT systems can also support condition monitoring when calibrated sensors, reliable communication and quality controls are available.

ConstructVision AI addresses this workflow problem by connecting inspection evidence to spatial context, transparent screening calculations, repair planning and reporting. The design principle is that a visual model, simulated sensor value or automated flag must never be presented as a certified engineering decision without appropriate validation and review.

3. Objectives

The primary objective is to design and demonstrate a civil-infrastructure inspection and site-intelligence prototype that helps organise evidence and supports qualified engineering review.

Specific objectives are:

- Develop a Streamlit-based workspace for inspection, asset context, condition review and reporting.
- Accept camera or uploaded inspection images and apply a transparent OpenCV-based feature-triage workflow.
- Estimate approximate visible-feature dimensions using a user-supplied pixel-to-millimetre calibration factor.
- Present a parametric 3D frame and structural-component context for inspection discussion.
- Generate synthetic telemetry for demonstration and define typed telemetry records for future integration.
- Calculate transparent preliminary indicators for gravity, flexure, wind pressure and hydrostatic pressure.
- Provide repair recommendations, illustrative cost estimation and report/history workflows.
- Preserve the distinction between implemented prototype features, synthetic demonstrations, readiness-only integrations and proposed future capabilities.

4. Scope of Work

The project focuses on an AI-assisted civil-infrastructure inspection workflow for residential and similar built assets. It provides a user interface through which an inspector or student can view project context, upload or capture an image, review visible features, view a parametric structural model, inspect demonstration telemetry and create report-ready information.

The current prototype will

- Accept field photographs, camera frames or a synthetic demonstration target.
- Apply grayscale/HSV processing, optional glare/vegetation suppression, Canny edge processing and contour-based feature filtering.
- Display annotated image features with approximate dimensions, rule-based severity and illustrative repair information.
- Display a parameterised 3D building/frame view and sensor-ready node context.
- Generate deterministic synthetic temperature, wind, strain and tilt data for interface demonstration.
- Perform preliminary, transparent screening calculations and clearly flag when review is required.
- Provide illustrative cost and bill-of-quantity style calculations.
- Maintain prototype history/report screens and define structured contracts for future asset, capture, telemetry and analysis records.

- Show external-worker readiness as READY or BLOCKED without fabricating a reconstruction, FEA or CFD result.

Limitations

- The image workflow is rule-based OpenCV triage; it is not a trained or validated YOLO model.
- Image dimensions are approximate and depend on the supplied calibration factor, camera position, surface condition and lighting.
- The interface can display synthetic telemetry; it does not demonstrate a production sensor gateway or validated real-time alert service.
- Structural calculations are preliminary screening indicators and are not code-compliant structural design checks, safety certificates or approvals.
- Internal damage cannot be confirmed from surface images alone; methods such as GPR, ultrasonic testing or other specialist investigation may be required.
- Validated reconstruction, FEA, CFD, seismic response, carbon accounting, authentication, immutable storage and role-based approvals are outside the current internship prototype scope.

5. Plan of Work

Phase	Duration	Main activities
Phase 1	Week 1	Requirement analysis, literature review, project abstract, environment setup and repository review.
Phase 2	Week 2	Image-workflow development, parameterised twin development, data/contract design and initial interface implementation.
Phase 3	Week 3	Screening and cost workflow integration, dashboard/report development and feature testing.
Phase 4	Week 4	Documentation, verification of implemented/proposed claims, presentation preparation and prototype demonstration.

6. Literature References

- A. A. Elhariri et al., "Computer vision framework for crack detection of civil infrastructure: A review," *Journal of Building Engineering*, vol. 67, 2023, doi: 10.1016/j.jobbe.2022.105979.
- S. Dorafshan, R. J. Thomas and M. Maguire, "Comparison of deep convolutional neural networks and edge detectors for image-based crack detection in concrete," *Construction and Building Materials*, vol. 186, pp. 1031-1045, 2018.
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- C. R. Farrar and K. Worden, *Structural Health Monitoring: A Machine Learning Perspective*. Wiley, 2012.
- OpenCV, "OpenCV documentation." <https://docs.opencv.org/>
- Streamlit, "Streamlit documentation." <https://docs.streamlit.io/>

7. Particulars of the Supervisor

Particular	Details
Name	Ms. Swati P. Maniyal
Qualification	M.TECH, Structural Engineering
Designation	College Guide
Industry guide	Mr. Akash S. Chatake, M.TECH, AIML, BITS Pilani

Particular	Details
Organisation	Chatake Innworks Pvt. Ltd., MindForgeAI Division

8. Remarks of the Supervisor

Supervisor remarks:

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Date of review	Student signature	Supervisor signature